PHARMACOLOGY LAB 2(2026-27)

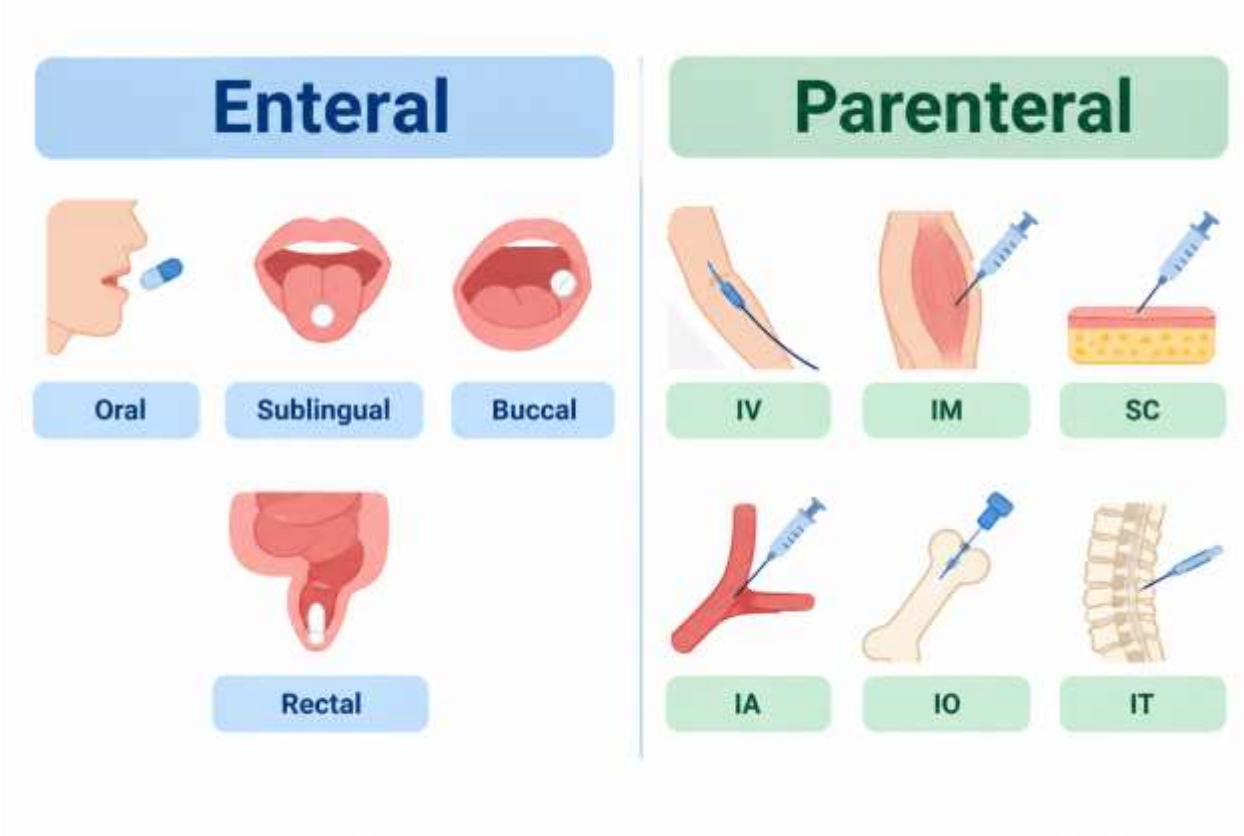
EFFECT OF ROUTES OF DRUG ADMINISTRATION ON DRUG RESPONSE

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ROUTES OF DRUG ADMINISTRATION

- There are two major routes of drug administration ***enteral*** and ***parenteral***



The choice of the route is determined by many factors such as:

1. Patient state (as in an emergency state, conscious or unconscious)
2. Onset of action .
3. Drug characteristic (water or lipid solubility….. etc).
4. Desired response (systemic or local effect).

A/ ENTERAL ADMINISTRATION

The enteral routes are the method in which the drug is absorbed from the gastrointestinal tract (GIT) these include.:

1. ***Oral***
2. ***Sublingual***
3. ***Buccal and***
4. ***Rectal routes.***

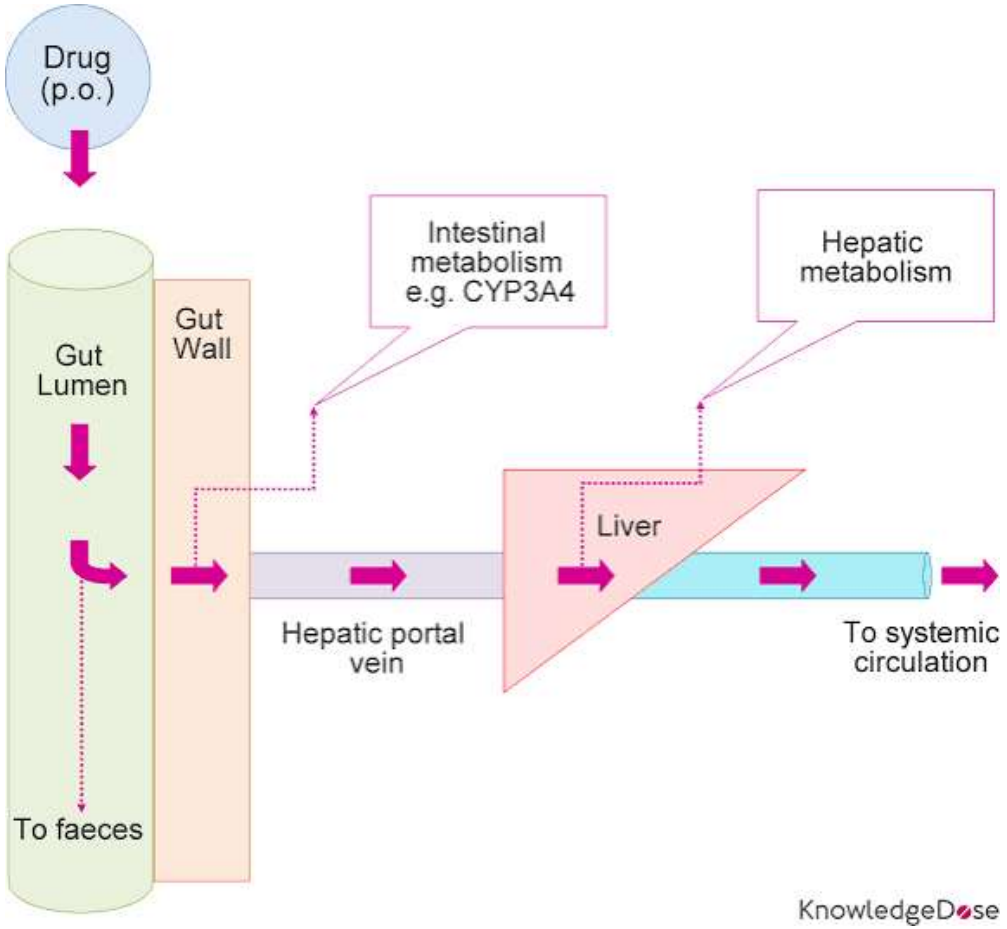
I. ORAL:

- The medication is swallowed by mouth and drug is absorbed from the stomach and small intestines. It is most common route of administration, it has some advantages and disadvantages .
- The advantages:
- A-economical
- B-convenient
- C-relatively safe

Disadvantages:

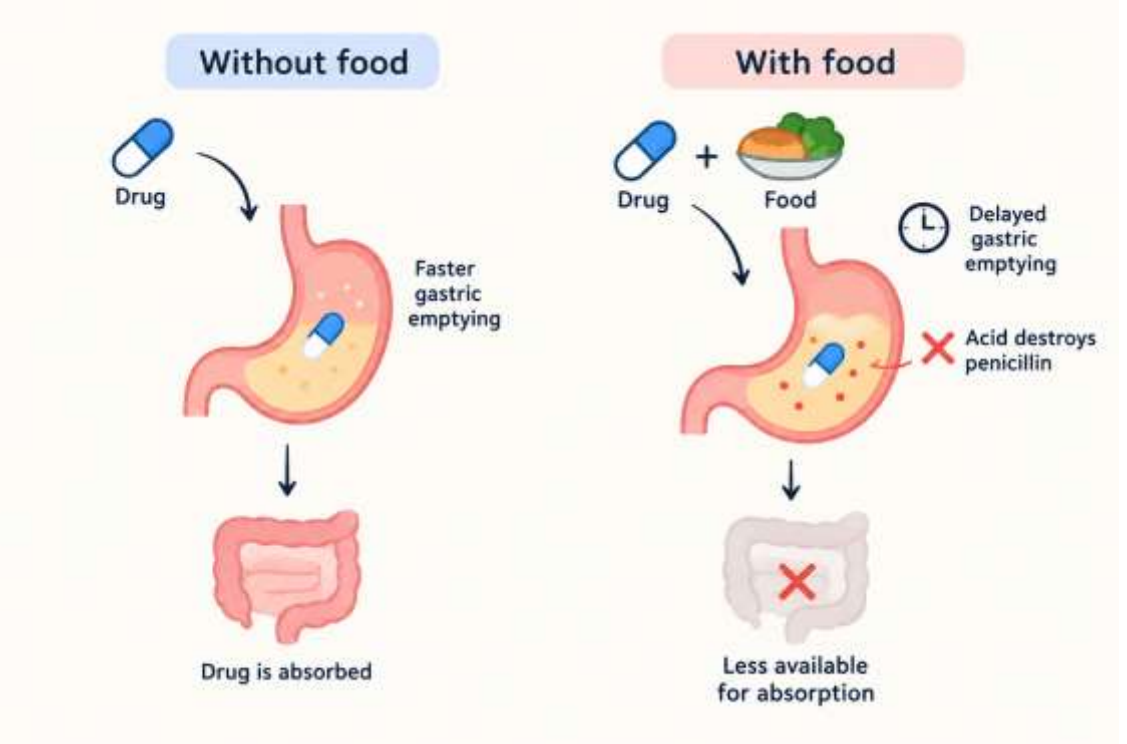
a- The most complicated pathway to the tissues

b- First-pass metabolism by the liver limit efficiency of many drugs. (Most drugs absorbed from GI tract enter **the portal circulation** and encounter the liver before distributed in the general circulation).



c- Ingestion of drugs with food can influence absorption. The presence of food in the stomach delays gastric emptying time so that the drugs are destroyed by acid (for example penicillin become unavailable for absorption).

Enteric coating of drug protect it from acid, ***sustained-release*** prolong the release of the drug.



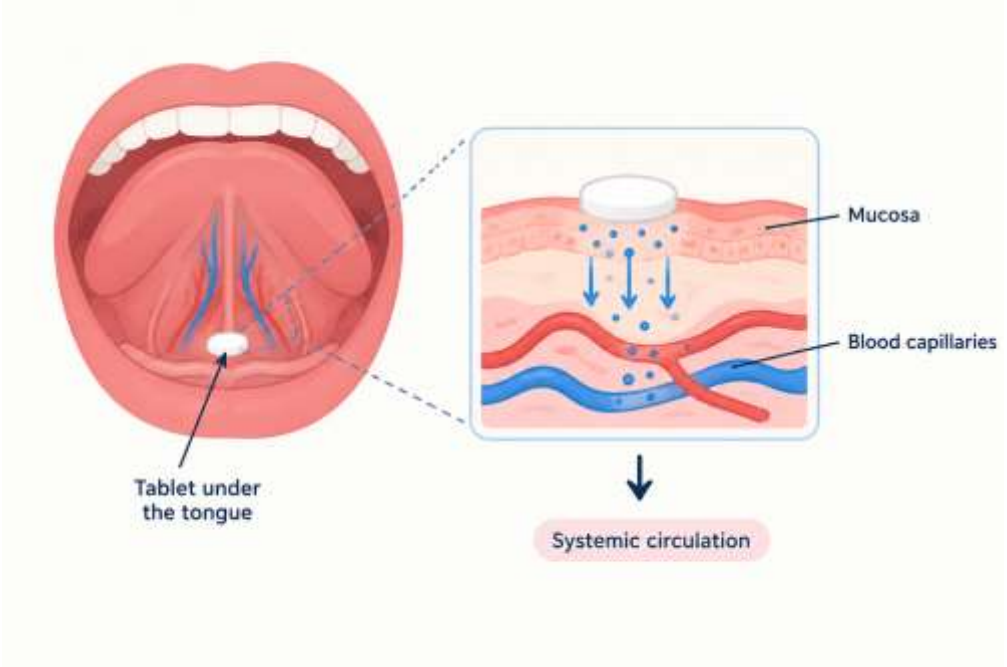
d- It is not suitable for use by patients who are sedated, comatose or suffering from vomiting.

e- Slow and incomplete absorption.

f- Other factors such as gastrointestinal disease and other medicines

1. SUBLINGUAL ROUTE:

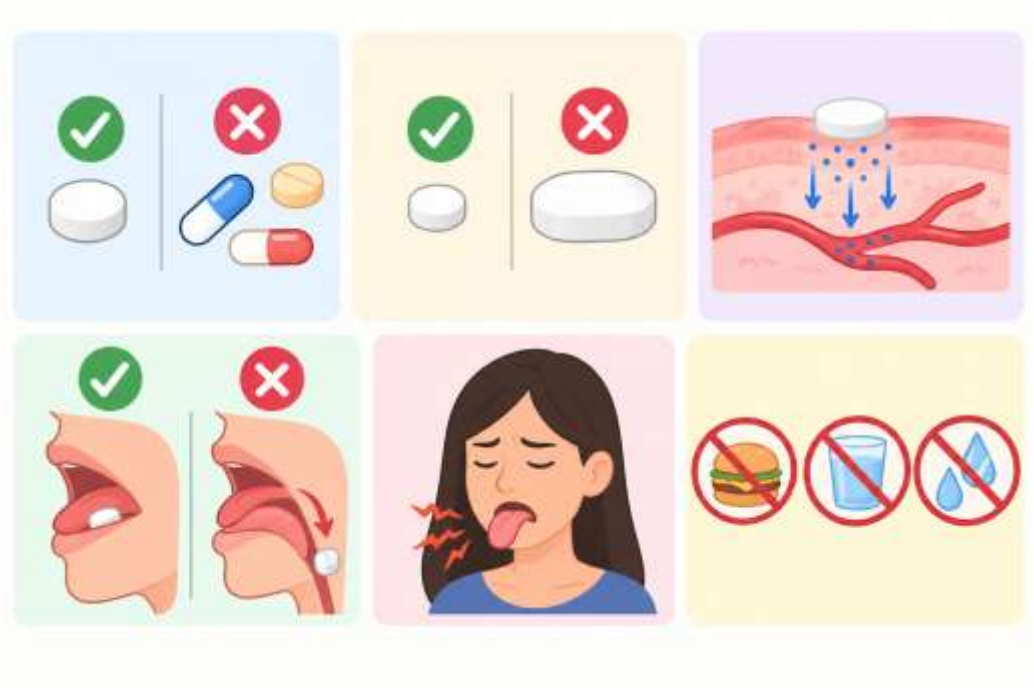
- The product is placed under the tongue; this allows the drug to diffuse into the capillary network and therefore it enters the systemic circulation directly.



1. SUBLINGUAL ROUTE:

- The advantage of this route is the rapid absorption of certain drugs from this site of administration to avoid first-pass drug metabolism.

- Disadvantage?



1. BUCCAL ROUTE:

- In the buccal route of administration, the drug is placed between the cheek and the gum, this method has the same advantages and disadvantages as the sublingual route,
- Two examples of drug available (nitroglycerine for treating ischemic heart disease and hyoscyamine for treating bowel cramp).







1. RECTAL ROUTE:

- The administration of drug is as suppository form inserted in to the rectum, this route is used either to obtain localized effect or systemic effect.



- Suppositories are useful when patients cannot take medication by mouth, or for treating localized condition such as hemorrhoids.



- Drugs absorbed from the rectum undergo relatively little first-pass metabolism in the liver.

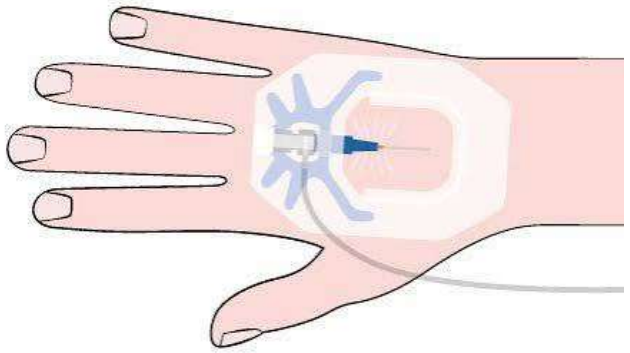


B/PARENTERAL ADMINISTRATION

- This method is used for drugs that are **poorly absorbed from the GIT**, and for agents that are **unstable in GIT** such as insulin.
- Parenteral route is used for treatment of unconscious patients and Under circumstances that require a rapid onset of action;
- This way provides the **most control** delivering the actual dose of drug to body
- Faster and more complete absorption than oral route.

I. INTRAVASCULAR ROUTE (IV)

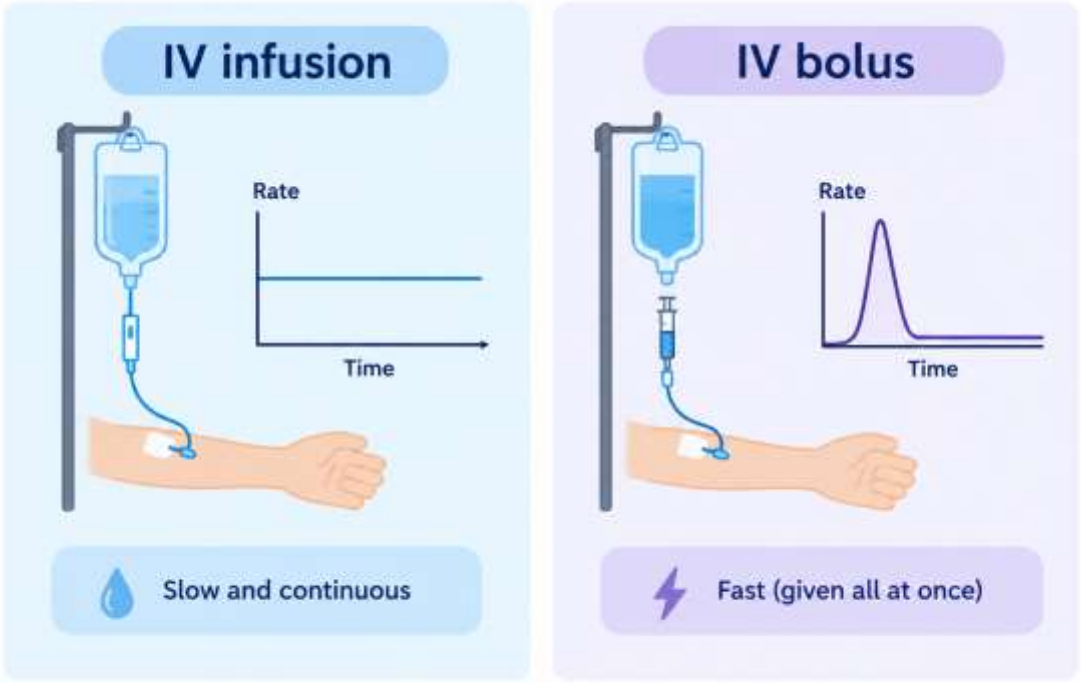
- Intravenous IV injection is the most common parenteral route
- The drug is administrated into the vein by mean of needle and syringe or infusion set.



I. INTRAVASCULAR ROUTE (IV)

Advantage:

- With IV. administration the drug will bypass the GIT therefore first-pass metabolism is avoided.
- This route permits a rapid effect.



I. INTRAVASCULAR ROUTE (IV)

Disadvantages: such as ***contamination with bacteria*** or adverse reaction by too rapid delivery of high concentration of drug to the plasma and tissues.

Therefore, the rate of infusion must be carefully controlled.

Similar concerns apply to **intra-arterial IA** Injected drugs, this route is very limited method used with drugs.



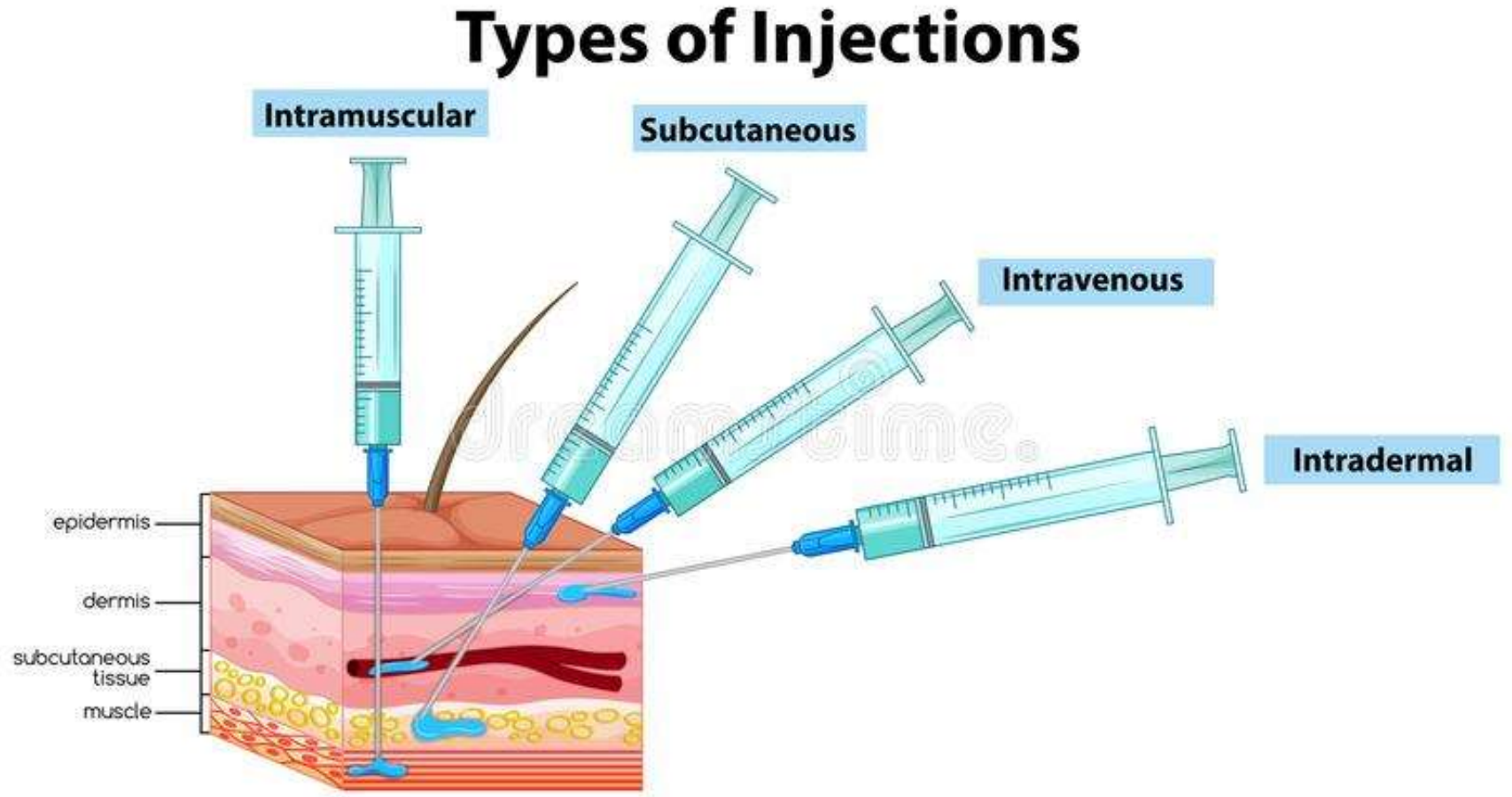
2-INTRA-MUSCULAR ADMINISTRATION I.M.:

- The drug is injected into the gluteal region where the drug is absorbed rapidly because the circulation of blood to this muscle .



3-SUBCUTANEAOUS ADMINISTRATION S.C.:

- The drug is injected beneath the skin so slower absorption than IM.



4-INTRATHECAL ADMINISTRATION



Refers to injection of a drug through the theca of the spinal cord and into subarachnoid space (for example in cases of meningitis). It is useful in administrating antibiotics that do not cross the blood brain barrier.

5. INTRA-ARTICULAR:



Administration into the joints as in it's used to treat arthritis.

6. INTRA-PERITONEAL (IP):



C/OTHER ROUTES

1. Transdermal:

- This route of administration can achieve **systemic effects** by application of drugs to the skin usually via **transdermal patch**.
- This route is often used for the **sustained delivery of drugs** such as anti-angina drug, Nitroglycerin highly soluble in lipid membrane.
- Absorption is slow, but first- pass metabolism is avoided

2- INHALATION:

- Inhalation provides the rapid delivery of a drug across the large surface area of the mucous membrane of the respiratory tract and pulmonary epithelium producing either a localized or systemic drug effect.
- This route is particularly effective and convenient for patients with respiratory complains (for example asthma or chronic obstructive pulmonary diseases), systemic side effects are minimized.

3- INTRANASAL:

- The nasal spray form of the drug is used intranasally for the treatment of some diseases either systemic or local effect*
- Example of systemic such as Desmopressin and calcitonin,
- Example of local Beconase spray*
- The abused drug, cocaine is generally taken by sniffing.

4- TOPICAL:

- Application of drugs to the surface of the body to produce a localized effect, it's often used for treating disorders of skin, eyes and ear.

5/ VAGINAL:

- in Gynecology for the treatment of female reproductive system diseases.

EXPERIMMENT

- **Objective**

When a drug is administered by a route other than intravascular, it will be absorbed from site of administration to the site of action to produce the desired response. This process will affect the onset of action of the drug (response).

Aim of the experiment:

The purpose of this laboratory experiment is to observe how the ***onset of response*** and ***duration of response*** to a drug differs when administered by different routes.

Experimental Procedure:

- Drug: Ketamine
- Laboratory animal: rabbit
- Students are divided to two groups:
- Group1: administer ketamine IP to a rabbit
- Group2: administer ketamine IM to another rabbit.

- **Ketamine:**

is an anesthetic agent administered parenterally to produce sedation and anesthesia for surgical operation

- Investigations include recording time of:

1- Onset of response.

2- Duration of response

Response: sedation, unconsciousness

IP Route		IM route	
Onset of Response (minutes)	Duration of response (minutes)	Onset of Response (minutes)	Duration of response (minutes)

